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The Ulvö Gabbro Complex of the 1.27–1.25 Ga Central Scandinavian Dolerite Group (CSDG): Intrusive age, magmatic setting and metamorphic history

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Abstract: The Ulvö Gabbro Complex (UGC) belongs to the 1.27–1.25 Ga Central Scandinavian Dolerite Group (CSDG), which manifests a major event of intra-cratonic mafic magmatism possibly related to break-up between Laurentia and Baltica. Lopolithic, layered intrusions of the UGC outcrop in east central Sweden and west central Finland. Fresh baddeleyite grains in a pegmatoidal gabbro were dated by the TIMS U–Pb method. The age of 1256.2±1.1 Ma is interpreted to date crystallisation of the UGC, and is consistent with the age of other mafic intrusions of the CSDG in Västerbotten and south-western Finland. Previously reported K–Ar biotite data confirm that the UGC was not heated above the blocking temperature for Ar diffusion in biotite (~250˚C) after ~1.2 Ga. This suggests that UGC was virtually unaffected by later geological events and that magmatic textures and mineralogy have not been modified after solidification.

Keywords: Ulvö Gabbro Complex, ulvöspinel, baddeleyite, U–Pb, CSDG, layered intrusion.

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Introduction

The Ulvö Gabbro Complex (UGC; Fig. 1A) has been the subject of geological research for more than 150 years, and is the type locality of the common oxide mineral ulvöspinel (Mogensen 1946). This suite of intrusions has attracted attention because of the well-developed igneous layering in excellent coastal exposures (Fig. 2), and the economic concentrations of Fe-Ti oxides (Larson 1980; Lundqvist et al. 1990). The UGC belongs to the late Mesoproterozoic (~1.26 Ga) Central Scandinavian Dolerite Group (CSDG; Fig. 1B), which has been suggested to be associated with the break-up of Baltica and Laurentia (Patchett & Bylund 1977; Patchett et al. 1978; Gorbatschev et al. 1979; Elming & Mattson 2001). The CSDG consists of five major complexes; Dalarna, Jämtland, Västerbotten, Satakunta, and Ulvö that cover an area of at least 100 000 km^2 in the Svecofennian part of Sweden and Finland (Elming & Mattson 2001). The sills and dykes of the CSDG have alkali basaltic compositions, (Fig. 3; Lundqvist & Samuelsson 1973; Gorbatschev et al. 1979; Larson 1980; Aro 1987; Hämäläinen 1987; Solyom et al. 1992; Patchett et al. 1994) and recent advances in chronology demonstrate intrusion in three separate pulses at 1271–1264 (Dalarna), 1259–1256 (Västerbotten, Satakunta) and 1247–1246 Ma (Jämtland; Söderlund et al. 2004b; 2005b; subm.).

Previous age constraints of the UGC include 1215±20 Ma (K–Ar biotite-WR) at Långroudden, Sweden (Welin & Lundqvist 1975; Welin 1980), 1268±13 Ma (U–Pb zircon) at Norrgrynnan, Finland (Suominen 1991) and 1257.4±4.3 Ma (U–Pb baddeleyite) at Alnö, Sweden (Söderlund et al. subm). In this study we present a new intrusive age of this complex by TIMS U–Pb dating of baddeleyite that places the UGC in the 1259–1256 pulse of CSDG. Furthermore, the new age is used to discuss the origin of the voluminous UGC magmas and to constrain upper limit metamorphic conditions. This information will provide an essential framework for future studies of magmatic processes in the UGC.

Geological setting and field relations

The UGC was emplaced in the Svecofennian Bothnian basin that comprises 1.95–1.87 Ga metagreywackes and several generations of granitoids and subordinate gabbros (1.89 to 1.59 Ga;

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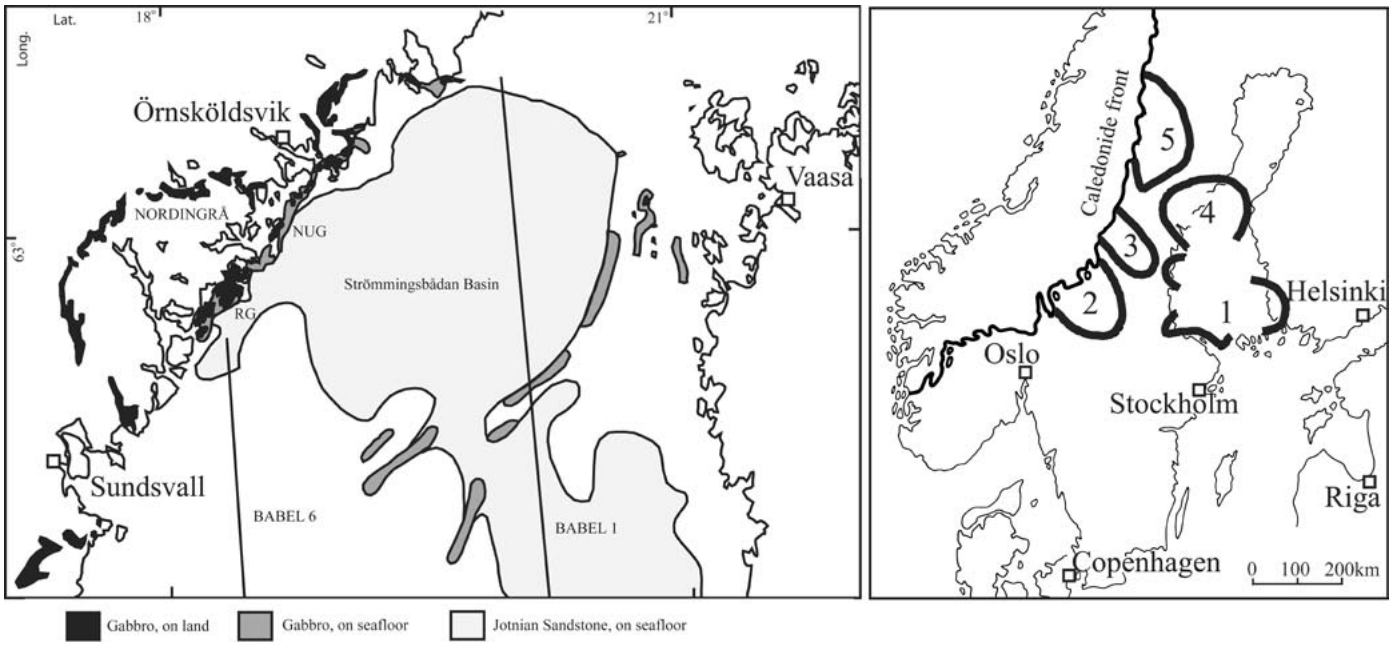


Fig. 1. **A.** Map of the Ulvö Gabbro Complex, which is re-drawn from the Geological map of the Fennoscandian Shield (Koistinen et al. 2001; NUG – Norra Ulvön Gabbro; RG – Ringkallen Gabbro). **B.** The regional extension of the Central Scandinavian Dolerite Group modified after Gorbatschev et al. (1979) and Elming & Mattson (2001). The complexes are 1. Satakunta, 2. Dalarna, 3. Jämtland, 4. Ulvö and 5. Västerbotten.

Welin & Lundqvist 1984; Claesson & Lundqvist 1995; Öhlander & Romer 1996; Lundqvist et al. 1998; Hellström 2004). The sills also intrude the ~3–4 km thick Jotnian sandstones of the Strömmingsbådan basin (Fig.1; Korja et al. 2001). The age of these sediments is poorly constrained, but similar Jotnian sandstones in west central Sweden (Dala-sandstones) have a minimum depositional age of ~1.46 Ga for the lower units (Söderlund et al. 2004a).

The UGC consists of a series of lopolithic intrusions stretching from the Nordingrå area on the Swedish east coast across the Bothnian basin to the Vaasa archipelago in western Finland (Winterhalter 1972; Larson 1980; BABEL Working Group 1993; Korja et al. 2001). Extrapolations of outcrops and geophysical data show a large circular gabbro structure around the Strömmingsbådan basin (Fig. 1A; Korja et al. 2001). In the Nordingrå area, outcrops of one big intrusion with a diameter of about 90 km and several smaller circular intrusions with diameters of 10–30 km have been mapped (Lundqvist et al. 1990). The gabbro sheets in this area dip gently (~10–20°) toward the centres of the circular structures located in the Baltic Sea (Larson 1980). Larson & Magnusson (1979) were able to follow these dipping sheets at depth by gravimetric measurements. The lopoliths can be seen as strong reflectors in the seismic data from the northern parts of Babel Line 1 and 6 (Fig. 1; Korja et al. 2001), where steeply dipping marginal parts can be correlated with outcrops of alkali gabbros on islands and on the floor of the Baltic Sea (Winterhalter 1972).

The thickness of the sheets range between 200 and 500 m in the Nordingrå area (Lundqvist & Samuelsson 1973; Larson 1980), but some sheets may be up to 1 km thick (Lundqvist 1973). Korja et al. (2001) estimated that the sills beneath the Strömmingsbådan basin are ~500–1000 m thick, but noted considerable thinning in the shallow marginal parts.

Depth of emplacement

An emplacement depth of ~3 km has been estimated for the gabbros in the Nordingrå area (Lundqvist & Samuelsson 1973; Larson 1980; Lundqvist et al. 1990). This is inferred from low temperatures in the country rock at the time of intrusion, as indicated by aphanitic, sometimes glassy chilled margins and the occurrence of columnar jointing in the marginal parts of the gabbros. The fact that many of the gabbros in this area have been emplaced in the Jotnian sandstone (~3–4 km thick) gives further support for emplacement at a shallow crustal level. Seismic reflectors interpreted as mafic sills can be seen as deep as 25 km down in the crust and all the way to the surface, but the majority of sills are found at a depth of around 10 km (Korja et al. 2001).

Geochemistry and mineralogy

Stratigraphic sections through the Norra Ulvö Gabbro (NUG) and the Ringkallen gabbro are relatively complete and have been the subject of detailed petrological studies by Lundqvist & Samuelsson (1973), Larson (1980) and Larson et al. (subm.). Analyses of chilled margin samples from these gabbros show that the parental liquid has an alkali basaltic composition (Fig. 3A, B), and both major and trace element data points to a remarkably uniform composition of this liquid (Larson et al. subm.). Chondrite normalized spidergram (Fig. 3B) reveals negative Nb and Ta anomalies which are uncommon for alkali basalts, but



Fig. 2. Gently dipping, rhythmic modal layering from the Ulvö Gabbro Complex. Thickness of individual layers is ~10 cm.

are consistent with data from other parts of the CSDG (Patchett et al. 1994). Patchett et al. (1994) explain the anomalies by chemical interaction between asthenosphere-derived magmas and the lithospheric mantle. The magma scavenges incompatible elements from the lithospheric mantle, whereas Nb and Ta are retained in spinel.

On the basis of mineralogical and textural variations, the stratigraphy in the ~300 m thick intrusions can be subdivided into lower, central and upper zones. The lower and upper zones (50 to 100 m thick) crystallized from the contacts inward, and are characterized by relatively homogeneous, medium-grained gabbro with ophitic to subophitic texture. Their mineralogy consists of cumulus plagioclase and olivine, with interstitial clinopyroxene, Fe-Ti oxides, biotite, apatite and K-feldspar.

The central portion of the stratigraphy is defined by the occurrence of rhythmic modal layering and distinct plagioclase lamination (Fig. 2). Cumulates in the central zone formed by floor accumulation, and where they meet the upper zone a "sandwich horizon" is formed that contains the last liquid crystallized. Differentiation of the magma in the central zone is reflected by variations in mineralogy such as an abrupt increase in modal olivine from 20% in the lower zone up to 50% at the base of the central zone. The increase in olivine is closely followed by saturation of Fe-Ti oxides and the appearance of layers with up to 55 modal% of euhedral grains of magnetite-ulvöspinel (in NUG). Next, clinopyroxene becomes a cumulus phase whereas modal olivine decreases to 2–5%. Finally, the liquid saturates in apatite close to the sandwich horizon. Core compositions of plagioclase and olivine show strong stratigraphic variations as a result of progressive fractionation of the magma. A continuous decrease in anorthite and forsterite can be seen from the margins (An ~70 and Fo ~70) inward towards the sandwich horizon (An ~45 and Fo ~25; Larson et al. subm.).

Gabbroic pegmatoids are common in the UGC both as stratiform layers and as discordant late-stage bodies. The pegmatoids are coarse-grained and contain relatively large amounts of accessory minerals (Larson 1980).

Dating methods

Approximately 2 kg of a pegmatoidal gabbro was collected for obtaining high-density minerals for U–Pb geochronology. The pegmatoid forms a ~50 cm thick layer, close to the sandwich horizon in the NUG profile (coordinates in the Swedish national grid: 6996380/1645360). Standard crushing, grinding and magnetic separation techniques were employed to separate heavy minerals. The sample contained large rod-shaped baddeleyite grains (up to 200 μm in length) that were hand-picked under a microscope. The grains are light brown, perfectly crystalline with no signs of secondary alteration.

Dissolution, ion exchange chromatography and mass spectrometry were performed at the Laboratory of Isotope Geology (LIG) at the Swedish Museum of Natural History in Stockholm. The baddeleyite grains were transferred to mini teflon capsules and washed in 2–3 M HNO_3 on hot plate for ~10 min. The sam-

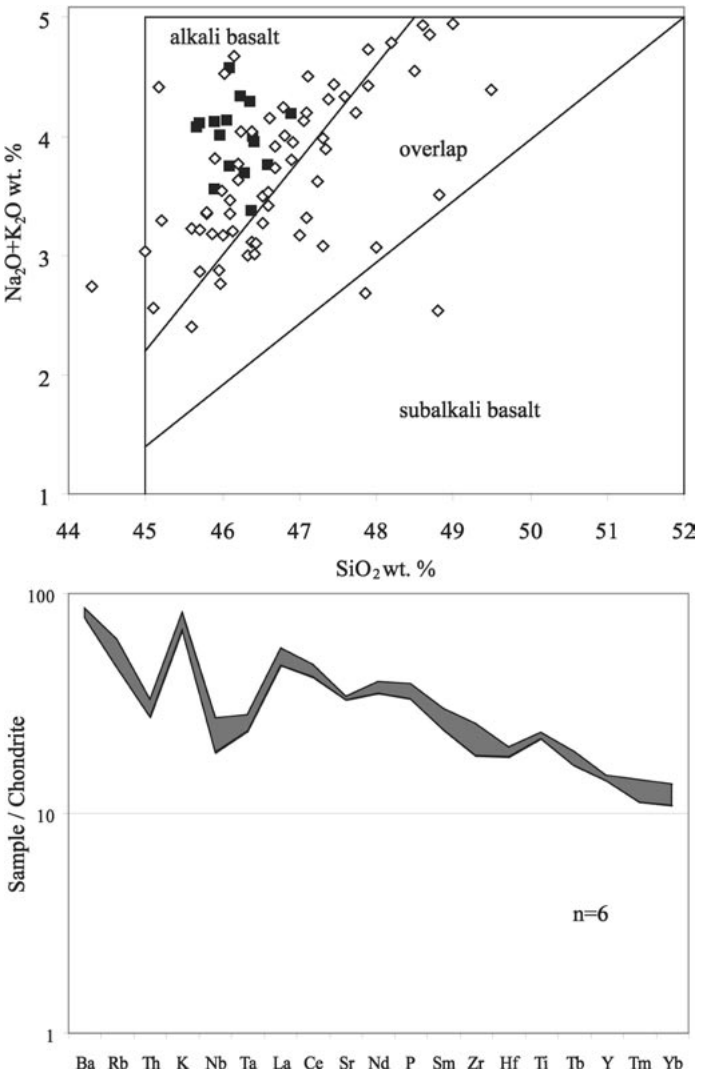


Fig. 3. **A.** Ulvö Gabbro Complex chilled margin samples (filled squares) plot in the field of alkali basalts of Bellieni et al. (1983) in the total alkali vs. silica diagram (Le Bas et al. 1986). Chemical analyses from other parts of the CSDG (diamonds; Lundqvist & Samuelsson 1973; Gorbatschev et al. 1979; Larson 1980; Aro 1987; Hämäläinen 1987; Solyom et al. 1992; Patchett et al. 1994; Elming & Mattson 2001) have been included for comparison. Most analyses point to an alkali basaltic composition of the CSDG magmas and according to Bellieni et al. (1983), the samples that plot in the "overlap field" are three times more likely to be alkali basalts than subalkali basalts. **B.** Chondrite normalized spiderdigram showing the range of contact sample trace element compositions from the UGC (Larson et al. subm.). The enriched character of these samples are typical for alkali-basalts.

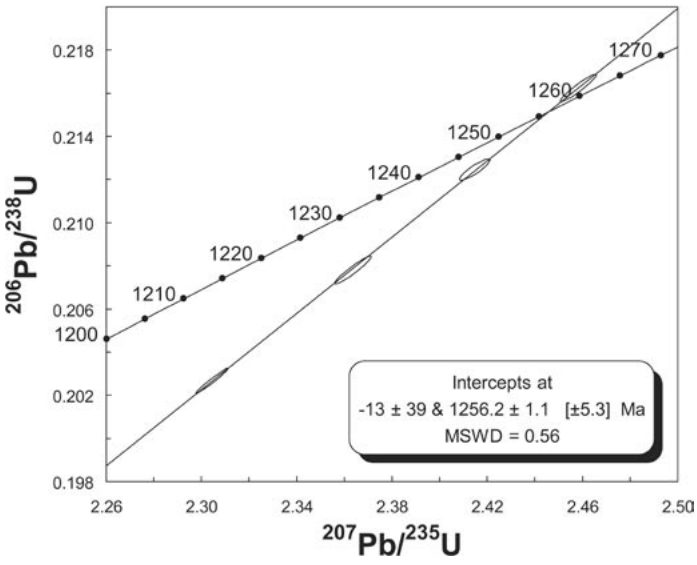


Fig. 4. The upper intercept of 1256.2±1.1 Ma in this concordia diagram is considered to represent the intrusive age of the Ulvö Gabbro Complex.

ples were thereafter washed several times in ultra-clean H_2O. ^{205}Pb-$^{233\text{-}236}U$ tracer and 10 drops of HF and HNO_3 (in proportions 10:1) were added to the capsules. The baddeleyite grains were completely dissolved over night at 210°C. After evaporation on hot plate the sample was re-dissolved in 3.1 M HCl and loaded on 50 μl columns with anion exchange resin. After washing by 3.1 M HCl, uranium and lead were eluted by H_2O and dried down on hot plate. The sample was loaded on a single Re filament with silica gel. Mass spectrometry analysis was performed on a Finnigan MAT 261. ^{208}Pb, ^{207}Pb, ^{206}Pb and ^{205}Pb were measured using Faraday collectors. The $^{207}Pb/^{204}Pb$, used for common lead correction, was measured in dynamic collector mode using a Secondary Electron Multiplier. After completed Pb isotopic measurements (in the 1200–1320°C range) the U isotopic composition was analyzed in dynamic multicollector mode at filament temperatures of ≥1350°C. Procedural blank is estimated to about 3 pg Pb and <1 pg U. Mass fractionation is 0.1% per mass unit for Pb, determined by replicate analyses of NBS standards SRM 981 and SRM 983. U fractionation was determined directly from the measured $^{233}U/^{236}U$ isotopic ratio. Initial Pb composi-

Table 1. U–Pb TIMS data.

Sample	Mineral type[1]	U/Th	Pbc/Pbtot[2]	$^{206}Pb/^{204}Pb$ raw[3]	$^{207}Pb/^{235}U \pm 2\sigma$% err [corr][4]		$^{206}Pb/^{238}U \pm 2\sigma$% err		$^{207}Pb/^{235}U$ age (Ma)	$^{206}Pb/^{238}U$ age (Ma)	$^{207}Pb/^{206}Pb$ age (Ma)
Ulvö	B1	57.2	0.0013	14494	2.4148	0.22	0.21246	0.19	1247.0	1241.9	1255.8
	B2	62.5	0.0008	27969	2.2948	0.28	0.20182	0.28	1210.7	1185.1	1256.6
	B3	38.6	0.0039	10022	2.4583	0.25	0.21624	0.24	1259.9	1262.0	1256.2
	B4	62.8	0.0009	18459	2.3636	0.27	0.20781	0.25	1231.7	1217.1	1257.2

[1] B – baddeleyite. All grains are clear and dark brown. About 20–40 grains/fragments were combined in each analysis.
[2] Pbc – common Pb; Pbtot – total Pb (radiogenic + blank + initial).
[3] measured ratio, corrected for fractionation and spike.
[4] isotopic ratios corrected for fractionation (0.06% per amu for Pb), spike contribution, blank (2–5 pg Pb and <1 pg U), and initial common Pb. Initial common Pb corrected with isotopic compositions from the model of Stacey and Kramers (1975) at 1256 Ma.

tions were taken from the model of Stacey & Kramers (1975) at 1260 Ma. Pb blank composition used was: $^{206}Pb/^{204}Pb$ = 18.5, $^{207}Pb/^{204}Pb$ = 15.6 and $^{208}Pb/^{204}Pb$ = 38.5. The decay constants for ^{238}U (1.55125×10^{-11}) and ^{235}U (9.8485×10^{-10}) are those recommended by Steiger & Jäger (1977). The listed uncertainties in Pb/U ratios (Table 1) were calculated by propagating the within-run error for measured isotopic ratios, and the uncertainties in fractionation (±0.04 for Pb, absolute uncertainties), in Pb and U blank concentration (±50%), and in the Pb blank composition (2% for $^{206}Pb/^{204}Pb$ and 0.2% for $^{207}Pb/^{204}Pb$).

Results

U–Pb data is listed in Table 1 and the age result is shown in concordia diagram in Fig. 4. Two fractions lay close the concordia curve, one of them falling slightly above the curve. The other two fractions are variable discordant indicating varying degree of Pb-loss. Together they define a discordia line that yields an upper intercept of 1256.2±1.1 Ma, taken as the age of the sample (MSWD = 0.6). The lower intercept is 0 Ma within error and indicates variable degree of modern Pb-loss. Our result is the most precise age determination performed on mafic intrusions belonging to the UGC.

Discussion

Magmatic setting

It is reasonably well known that the alkali gabbros in the Nordingrå area and Vaasa archipelago are parts of the same magmatic complex (UGC; e.g. Larson 1980; Korja et al. 2001) and the age of 1256.2±1.1 Ma is therefore interpreted as the age of the whole Ulvö Gabbro Complex. Our result is in perfect agreement with other recently dated dolerites in the Ulvö (2 sample), Satakunta (3 samples) and Västerbotten complexes (2 samples) that give consistent dates between 1256 and 1259 Ma (Söderlund et al. 2004b; subm.). Suominen (1991) reports three U–Pb ages (zircon and baddeleyite) of dolerites that may suggest the presence of a slightly older generation (~1265 Ma) of CSDG in south-western Finland. Those dates are, however, associated with relatively large errors, and cannot be conclusive in this regard.

On the basis of virtually identical paleomagnetic directions of CSDG, Elming & Mattsson (2001) suggested only small age differences between dolerites in different complexes. In light of recent age data demonstrating a ~25 Myr total age span between the oldest and youngest magmatic pulse an alternative explanation invoke a near-stationary position of Baltica in the 1271–1246 Ma time period or only longitudal movements (cf. Söderlund et al. subm.).

According to Ernst & Buchan (2001), the CSDG may relate to large-scale mantle plume activity contemporaneous with extensive plume-associated magmatism in North America and Greenland, e.g. the 1267 Ma Mackenzie dikes and the 1284–1279 Ma mid-Gardar dikes (Cadman et al. 1993; Upton et al. 2003). The reconstruction of Baltica and Laurentia at ~1.27 Ga is based on well defined key paleopoles for CSDG and the Mackenzie dike swarm (Buchan et al. 2000; Elming & Mattsson 2001). In these reconstructions the nearest CSDG is located at a distance of about 5000 km from the assumed focal point from which the Mackenzie dikes radiate. This suggests that CSDG, if associated with plume activity, originated from a plume different to those causing extensive plume magmatism in Laurentia (cf. Ernst & Buchan 2002; Söderlund et al. subm). Söderlund et al. (subm.) discuss subduction and protracted magmatism associated with plume tail activity as alternative mechanisms for the origin of CSDG.

Metamorphic history

Three different geological events could potentially have affected the Nordingrå-Vaasa area after the emplacement of the UGC. (1) The ~1.0 Ga (cf. Bingen et al. 2001) Sveconorwegian orogeny affected the crust of south-western Scandinavia. The uplift and erosion of this region resulted in an 8 km thick sediment cover in the south-eastern and central part of Sweden (Larson et al. 1999). The burial metamorphism reached prehnite-pumpellyite facies (Nyström & Levi 1980; Wickman et al. 1983) and caused isotopic disturbance of the Rb–Sr isotopic system (Verschure et al. 1987). (2) A more local event occurred at ~580–550 Ma when the southern part of the Nordingrå area was intruded by carbonatitic and alkaline magmas of the Alnö complex (Kresten et al. 1977; Brueckner & Rex 1980; Andersen 1996). The country rock was affected by varying degree of fenitisation, but the alteration seems to be restricted to the area around Sundsvall (Fig. 1; Lundqvist et al. 1990). (3) The Devonian foreland basin of the Scandinavian Caledonides was filled by >2.5 km thick piles of sediments that extended all the way into Finland. Apatite fission track ages of 402–317 Ma (Cederbom et al. 2000) in the Nordingrå area suggest that the foreland basin was covered by sediments for up to a 100 Ma (see also Söderlund et al. 2005a).

The gabbros of the UGC have well preserved igneous textures and mineralogy, indicating that these later events had little effect. However, minor surficial alteration occurs near fractures where plagioclase was transformed to sericite or epidote and olivine to iddingsite. Other alteration minerals (e.g. prehnite, hornblende) are found in marginal zones and in some gabbroic pegmatoids (Lundqvist et al. 1990) and could potentially have a metamorphic origin. The K–Ar biotite age of 1215±20 Ma (Welin & Lundqvist 1975; Welin 1980) is reasonably close to the intrusive age of UGC and implies that these rocks never have been heated above ~230–255°C (the temperature at which Ar diffusion is blocked in biotite ; Lister & Baldwin 1996) after ~1.2 Ga. This rules out regional metamorphism as a mechanism for growth of hornblende, which instead is attributed to late stage magmatic processes. Futhermore, the baddeleyite dating shows no signs of ancient Pb-loss, which is consistent with low metamorphic temperatures.

Conclusions

We present a new intrusive age of 1256.2±1.1 Ma for the Ulvö Gabbro Complex that was determined by TIMS U–Pb dating of baddeleyite. This age places the UGC in a time of global magmatism, which may be related to a multiple-plume event preceeding break-up of Baltica and Laurentia. The unmetamorphosed character of UGC is supported by the relatively good correspondences between the U–Pb age and a previously reported K–Ar age. An upper limit for post-1.2 Ga heating, i.e. above ~230–255°C, is inferred from the closure temperature of Ar diffusion in biotite.

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References

Andersen, T., 1996: Sr, Nd and Pb isotopic data of the Alnö carbonatite complex. *Abstract volume. 22nd Nordic Geological Winter Meeting*, 11.

Aro, K., 1987: Diabases in the Vaasa archipelago and Petalothi, western Finland. *In* K. Aro & I. Laitakari (eds.): Diabases and other Mafic Dyke Rocks in Finland. *Geological Survey of Finland, Report of investigation 76*, 173–178.

BABEL Working Group, 1993: Integrated seismic studies of the Baltic Shield using data in the Gulf of Bothnia region. *Geophysical Journal International 112*, 305–324.

Bellieni, G., Justin, V., Le Maitre, R.W., Piccirillo, E.M. & Zanettin, B., 1983: Proposal for a division of the basaltic (B) field of the TAS diagram. *IUGS Subcommission on the Systematics of Igneous Rocks. Circ 38, Contr 102*, 1–23.

Bingen, B., Birkeland, A., Nordgulen, Ø. & Sigmond, E.M.O., 2001: Correlation of supracrustal sequences and origin of terranes in the Sveconorwegian orogen of SW Scandinavia: SIMS data on zircon in clastic metasediments. *Precambrian Research 108*, 293–318.

Brueckner, H.K. & Rex, D.C., 1980: K–Ar and Rb–Sr geochronology and Sr isotopic study of the Alnö alkaline complex, northeastern Sweden. *Lithos 13*, 111– 119.

Buchan, K.L., Mertanen, S., Park, R.G., Pesonen, L.J., Elming, S.Å., Abrahamsen, N. & Bylund, G., 2000: Comparing the drift of Laurentia and Baltica in the Proterozoic: the importance of key palaeomagnetic poles. *Tectonophysics 319*, 167–198.

Cadman, A.C., Heaman, L., Tarney, J., Wardle, R. & Krogh, T.E., 1993: U-Pb geochronology and geochemical variation within two Proterozoic mafic dyke swarms. *Canadian Journal Earth Science 30*, 1490–1504.

Cederbom, C., Larson, S.Å., Tullborg, E.-L. & Stiberg, J.-P., 2000: Fission track thermochronology applied to Phanerozoic thermotectonic events in central and southern Sweden. *Tectonophysics 316*, 153–167.

Claesson, S. & Lundqvist, T., 1995: Origins and ages of Proterozoic granitoids in the Bothnian Basin, central Sweden: isotopic and geochemical constraints. *Lithos 36*, 115–40.

Elming, S.Å. & Mattsson, H., 2001: Post Jotnian basic intrusions in the Fennoscandian Shield, and the break up of Baltica from Laurentia: a palaeomagnetic and AMS study. *Precambrian Research 108*, 215–236.

Ernst, R.E. & Buchan, K.L., 2001: Large mafic magmatic events through time and links to mantle plume heads. In R.E. Ernst & K.L. Buchan (eds.): Mantle Plumes: Their Identification Through Time. *Geological Society of America Special Paper 352*, 483–575.

Ernst, R.E &. Buchan, K.L., 2002: Maximum size and distribution in time and space of mantle plumes: evidence from large igneous provinces. *Journal of Geodynamics 34*, 309–342.

Gorbatschev, R., Solyom, Z. & Johansson, I., 1979: The Central Scandinavian Dolerite Group in Jämtland, Central Sweden. *Geologiska Föreningens i Stockholm Förhandlingar 101*, 177–190.

Hämäläinen, A., 1987: The Postjotnian diabases of Satakunta. In K. Aro & I. Laitakari (eds.): Diabases and other Mafic Dyke Rocks in Finland. *Geological Survey of Finland, Report of investigation 76*, 173–178.

Hellström, F.A., 2004: Emplacement age of the Kläppsjö gabbro; a mafic layered intrusion in the Bothnian basin, north central Sweden. *GFF 126*, 331–338.

Koistinen, T., Stephens, M.B., Bogatchev, V.,Nordgulen, Ø., Wennerström, M. & Korhonen, J., 2001: *Geological map of the Fennoscandian Shield, scale 1:2000000*. Geological Surveys of Finland, Norway, Sweden and the North West department of natural resources of Russia.

Korja, A. Heikkinen, P. & Aaro, S., 2001: Crustal structure of the northern Baltic Sea paleorift, *Tectonophysics 331*, 341–358.

Kresten, P., Printzlau, I., Rex, D.C., Vartiainen, H. & Woolley, A.R., 1977: New ages of carbonatitic and alkaline ultramafic rocks from Sweden and Finland. *Geologiska Föreningens i Stockholm Förhandlingar 99*, 62–65.

Larson, S.Å., 1980: *Layered intrusions of the Ulvö dolerite complex, Ångermanland, Sweden*. PhD thesis, Gothenburg University, Institution Geology. Publ., A 36. 213 pp.

Larson, S.Å. & Magnusson, K.-Å., 1979: A gravity investigation of the dolerite area on the coast of Ångermanland, Sweden. *Geologiska Föreningens i Stockholm Förhandlingar 101*, 1–16.

Larson, S.Å., Tullborg, E.-L., Cederbom, C. & Stiberg, J.-P., 1999: Sveconorwegian and Caledonain foreland basins in the Baltic Shield revealed by fission-track thermochronology. *Terra Nova 11*, 210–215.

Le Bas, M.J., Le Maitre, R.W., Streckeisen, A. & Zanettin, B., 1986: A chemical classification of volcanic rocks based on the total alkali – silica diagram. *Journal of Petrology 27*, 745–750.

Lister, G.S. & Baldwin, S.L., 1996: Modelling the effect of arbitrary *P - T - t* histories on argon diffusion in minerals using the MacArgon program for the Apple Macintosh. *Tectonophysics 253*, 83–109.

Lundqvist, T., 1973: Potash feldspar megacryst of a granite at Skagsudde, central Sweden. *Sveriges geologiska undersökning C 687*, 1–32.

Lundqvist, T. & Samuelsson, L., 1973: The differentiation of a dolerite at Nordingrå, central Sweden. *Sveriges geologiska undersökning C 692*, 1–62.

Lundqvist, T., Gee, D.G., Kumpulainen, R. & Karis, L., 1990: Explanation to bedrock map of Västernorrlands Region (in Swedish). *Sveriges geologiska undersökning Ba 31*, 175–198.

Lundqvist, T., Vaasjoki, M. & Persson, P.-O., 1998: U–Pb ages of plutonic and volcanic rocks of the Svecofennian Bothnian Basin, Central Sweden, and their implications for the Palaeoproterozoic evolution of the Basin. *GFF 120*, 357–63.

Mogensen, F., 1946: A ferro-ortho-titanate ore from Södra Ulvön. *Geologiska Föreningens i Stockholm Förhandlingar* 68, 578–588.

Nyström, J.-O. & Levi, B., 1980: Pumpellyite-bearing Precambrian rocks and post-Svecokarelian regional metamorphism in central Sweden. *Geologiska Föreningens i Stockholm Förhandlingar 102*, 37–39.

Öhlander, B. & Romer, R., 1996: Zircon ages of granites occurring along the Central Swedish Gravity Low. *GFF 118*, 217–25.

Patchett, P.J. & Bylund, G., 1977: Age of Grenville belt magnetisation: Rb-Sr and palaeomagnetic evidence from Swedish dolerites. *Earth and Planetary Science Letters 35*, 92–104.

Patchett, P.J., Bylund, G. & Upton, B.G.J., 1978: Palaeomagnetism and the Grenville orogeny: new Rb-Sr ages from dolerites in Canada and Greenland. *Earth and Planetary Science Letters 40*, 349–364.

Patchett, P.J., Lehnert, K., Rehkämper, M. & Sieber, G.,1994: Mantle and crustal effects on the geochemistry of Proterozoic dikes and sills in Sweden. *Journal of Petrology 35*,1095–1125.

Söderlund, P., Juez-Larré, J., Page L.M. & Dunai, T.J., 2005a: Extending the time range of apatite (U–Th)/He thermochronometry in slowly cooled terranes: Palaeozoic to Cenozoic exhumation history of southeast Sweden. *Earth and Planetary Science Letters 239*, 266–275.

Söderlund, U., Patchett, P.J., Isachsen, C.E. Vervoort, J.D. & Bylund, G., 2004a: Baddeleyite U–Pb dates and Hf–Nd isotope compositions of mafic dyke swarms in Sweden and Finland. *GFF 126*, 38.

Söderlund, U., Patchett, P.J., Vervoort, J.D. & Isachsen, C.E., 2004b: The ^{176}Lu decay constant determined by Lu-Hf and U-Pb isotope systematics of Precambrian mafic intrusions. *Earth and Planetary Science Letters 219*, 311–324.

Söderlund, U., Isachsen, C.E., Bylund, G., Heaman, L.M., Patchett, P.J., Vervoort, J.D. & Andersson, U.B., 2005b: U-Pb baddeleyite ages and Hf, Nd isotope chemistry constraining repeated mafic magamtism in the Fennoscandian Shield from 1.6 to 0.9 Ga. *Contributions to Mineralogy and Petrology 150*, 174–194.

Solyom, Z., Lindqvist, J.E. & Johansson, I.,1992: The geochemistry, genesis, and geotectonic setting of Proterozoic mafic dyke swarms in southern and central Sweden. *Geologiska Föreningens i Stockholm Förhandlingar 114*, 47–56.

Stacey, J.S. & Kramers, J.D., 1975: Approximation of terrestrial lead isotope evolution by a two-stage model. *Earth and Planetary Science Letters 26*, 207–221.

Steiger, R.H. & Jäger, E., 1977. Subcommision on Geochronology. Convention on the use of decay constants in geo- and cosmochronology. *Earth and Planetary Science Letters 36*, 359–362.

Suominen, V., 1991: The chronostratigraphy of southwestern Finland with special references to Postjotnian and Subjotnian diabases. *Geological Survey of Finland Bulletin 356*, 1–100.

Upton, B.G.J., Emeleus, C.H., Heaman, L.M., Goodenough, K.M. & Finch, A.A., 2003: Magmatism of the mid-Proterozoic Gardar ovince, South Greenland: chronology, petrogenesis and geological setting. *Lithos 68*, 43–65.

Verschure, R.H., Oen, I.S. & Andriessen, P.A.M., 1987: Isotopic age-determinations in Bergslagen, Sweden: VIII. Sveconorwegian Rb-Sr resetting and anomalous radiogenic argon in the Gothian TransScandinavian Småland-Värmland Granitic Belt and bordering parts of the Svecokarelian Bergslagen Region. *Geologie Mijnbouw 66*, 111–120.

Welin, E., 1980: Tabulation of recalculated radiometric ages published 1960–1979 for rock and minerals in Sweden. *Geologiska Föreningens i Stockholm Förhandlingar 101*, 309–320.

Welin, E. & Lundqvist, T., 1975: K–Ar ages of Jotnian dolerites in Västernorrland County, central Sweden. *Geologiska Föreningens i Stockholm Förhandlingar 97*, 83–88.

Welin, E. & Lundqvist, T., 1984: Isotopic investigation of the Nordingrå rapakivi massif, North-Central Sweden. *Geologiska Föreningens i Stockholm Förhandlingar 106*, 41–9.

Wickman, F.E., Åberg, G. & Levi, B., 1983: Rb-Sr dating of alteration events in granitoids. *Contributions to Mineralogy and Petrology 83*, 358–362.

Winterhalter, B., 1972: On the geology of the Bothnian Sea, a epeiric sea that has undergone Pleistocene glaciation. *Geological Survey of Finland Bulletin 258*, 1–66.